

1A)

$$U = U(T, V)$$

$$dU = \left(\frac{\partial u}{\partial T}\right)_V dT + \left(\frac{\partial u}{\partial V}\right)_T dV$$

$$\frac{dU}{dT}_V dT = C_V dT$$

$$C_V = T \left(\frac{\partial S}{\partial T}\right)_V$$

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T \frac{\partial^2 S}{\partial T \partial V}$$

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T \frac{\partial^2 P}{\partial T^2}$$

1B)

$$dS = Cp \frac{dT}{T} - R \frac{dP}{P}$$

$$Cp = A + B \frac{\left(\frac{C}{T}\right)^2}{\left(\sinh\left(\frac{C}{T}\right)\right)^2} + D \frac{\left(\frac{E}{T}\right)^2}{\left(\cosh\left(\frac{E}{T}\right)\right)^2}$$

$$0 = Cp \frac{dT}{T} - R \frac{dP}{P}$$

$$R \frac{dP}{P} = Cp \frac{dT}{T}$$

$$\frac{R \frac{dP}{P}}{\frac{T}{dT}} = Cp \quad \longrightarrow \quad Cp = R \frac{dP}{P} * \frac{T}{dT}$$

$$R \frac{dP}{P} * \frac{T}{dT} = A + B \frac{\left(\frac{C}{T}\right)^2}{\left(\sinh\left(\frac{C}{T}\right)\right)^2} + D \frac{\left(\frac{E}{T}\right)^2}{\left(\cosh\left(\frac{E}{T}\right)\right)^2}$$

$$\frac{dT}{T} \left(R \frac{dP}{P} * \frac{T}{dT} \right) = \left(A + B \frac{\left(\frac{C}{T}\right)^2}{\left(\sinh\left(\frac{C}{T}\right)\right)^2} + D \frac{\left(\frac{E}{T}\right)^2}{\left(\cosh\left(\frac{E}{T}\right)\right)^2} \right) \frac{dT}{T}$$

$$\int R \frac{dP}{P} = \int A \frac{dT}{T} + B \int \frac{\left(\frac{C}{T}\right)^2}{\left(\sinh\left(\frac{C}{T}\right)\right)^2} + D \int \frac{\left(\frac{E}{T}\right)^2}{\left(\cosh\left(\frac{E}{T}\right)\right)^2} \frac{dT}{T}$$

$$R \ln \left(\frac{P_2}{P_1} \right) = A \ln \left(\frac{T_2}{T_1} \right) + B \left[\frac{C \coth\left(\frac{C}{T}\right)}{T} - \ln * \sinh \left(\frac{C}{T} \right) \right] \Big|_{T_1}^{T_2} + D \left[\frac{-E \tanh\left(\frac{E}{T}\right)}{T} + \ln * \cosh \left(\frac{E}{T} \right) \right] \Big|_{T_1}^{T_2}$$

$$0 = (-R \ln \left(\frac{P_2}{P_1} \right)) + A \ln \left(\frac{T_2}{T_1} \right) + B \left[\frac{C \coth\left(\frac{C}{T}\right)}{T} - \ln(\sinh \left(\frac{C}{T} \right)) \right] \Big|_{T_1}^{T_2} + D \left[\frac{-E \tanh\left(\frac{E}{T}\right)}{T} + \ln(\cosh \left(\frac{E}{T} \right)) \right] \Big|_{T_1}^{T_2}$$

Using Secant method in excel, it was calculated that $T_2=117.6836$ K

$$f(0) = \left(-R \ln \left(\frac{P_2}{P_1} \right) \right) + A \ln \left(\frac{T_2(0)}{T_1} \right) + B \left[\frac{C \coth \left(\frac{C}{T_2(0)} \right)}{T_2(0)} - \ln \left(\sinh \left(\frac{C}{T_2(0)} \right) \right) \right] + D \left[\frac{-E \tanh \left(\frac{E}{T_2(0)} \right)}{T_2(0)} + \ln \left(\cosh \left(\frac{E}{T_2(0)} \right) \right) \right] - B \left[\frac{C \coth \left(\frac{C}{T_1} \right)}{T_1} - \ln \left(\sinh \left(\frac{C}{T_1} \right) \right) \right] + D \left[\frac{-E \tanh \left(\frac{E}{T_1} \right)}{T_1} + \ln \left(\cosh \left(\frac{E}{T_1} \right) \right) \right] +$$

$$f(1) = \left(-R \ln \left(\frac{P_2}{P_1} \right) \right) + A \ln \left(\frac{T_2(1)}{T_1} \right) + B \left[\frac{C \coth \left(\frac{C}{T_2(1)} \right)}{T_2(1)} - \ln \left(\sinh \left(\frac{C}{T_2(1)} \right) \right) \right] + D \left[\frac{-E \tanh \left(\frac{E}{T_2(1)} \right)}{T_2(1)} + \ln \left(\cosh \left(\frac{E}{T_2(1)} \right) \right) \right] - B \left[\frac{C \coth \left(\frac{C}{T_1} \right)}{T_1} - \ln \left(\sinh \left(\frac{C}{T_1} \right) \right) \right] + D \left[\frac{-E \tanh \left(\frac{E}{T_1} \right)}{T_1} + \ln \left(\cosh \left(\frac{E}{T_1} \right) \right) \right]$$

T2(0)	T2(1)	f(0)	f(1)	T2
173.15	153.15	3.072457	2.095309	110.2638
153.15	110.2638	2.095309	-0.51794	118.7638
110.2638	118.7638	-0.51794	0.072667	117.718
118.7638	117.718	0.072667	0.002322	117.6835
117.718	117.6835	0.002322	-1.1E-05	117.6836
117.6835	117.6836	-1.1E-05	1.55E-09	117.6836
117.6836	117.6836	1.55E-09	1.07E-14	117.6836
117.6836	117.6836	1.07E-14	6.75E-14	117.6836
117.6836	117.6836	6.75E-14	6.22E-14	117.6836
117.6836	117.6836	6.22E-14	1.4E-13	117.6836
117.6836	117.6836	1.4E-13	-3.6E-14	117.6836
117.6836	117.6836	-3.6E-14	-8.9E-16	117.6836
117.6836	117.6836	-8.9E-16	-5E-14	117.6836
117.6836	117.6836	-5E-14	-8.9E-16	117.6836
117.6836	117.6836	-8.9E-16	-8.9E-16	#DIV/0!
117.6836	#DIV/0!	-8.9E-16	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

(all constants were labeled in excel accordingly and were frozen when called in the function)

Secant Method: $x^{k-1} = x^k - \frac{f(x^k)}{f(x^k) - f(x^{k-1})} (x^k - x^{k-1})$

$$T2 = T_2(1) - \frac{f(1)}{f(1) - f(0)} * (T_2(1) - T_2(0))$$

1C)

$$\text{Sackur Tetrode Equation: } \frac{S}{kN} = \ln \left[\frac{V}{N} \left(\frac{4\pi m}{3h^2} \right) \frac{U}{N} \right]^{\frac{3}{2}} + \frac{5}{2}$$

$$U = U(S, V)$$

$$du = \left(\frac{dU}{dS} \right)_V dS + \left(\frac{dU}{dV} \right)_S dV = T dS - P dV$$

Maxwell Equations

$$\left(\frac{\partial U}{\partial S} \right)_V = T \quad \left(\frac{\partial U}{\partial V} \right)_S = -P$$

$$\left(\frac{\partial}{\partial V} \right)_S \left(\frac{\partial U}{\partial V} \right)_V = \left(\frac{\partial T}{\partial V} \right)_S$$

$$\left(\frac{\partial}{\partial S} \right)_V \left(\frac{\partial U}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

$$\left(\frac{\partial}{\partial S} \right)_V \left(\frac{\partial}{\partial V} \right)_S U = \left(\frac{\partial}{\partial V} \right)_S \left(\frac{\partial}{\partial S} \right)_V U$$

$$\left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

$$S = kN \left(\ln \left[\frac{V}{N} \left(\frac{4\pi m}{3h^2} \right) \frac{U}{N} \right]^{\frac{3}{2}} + \frac{5}{2} \right)$$

$$S = kN \left(\ln V + \ln \left(\frac{4\pi m}{3h^2} \right)^{\frac{3}{2}} \right) + \frac{5}{2}$$

$$\frac{\partial S}{\partial V} = * \frac{1}{V} kN \left(\frac{\partial}{\partial V} + \left(\frac{4}{3} \right)^{\frac{3}{2}} \right) + \frac{5}{2}$$

$$\frac{\partial U}{\partial S} = \frac{1}{V} kN \frac{\partial}{\partial V} \left(\frac{\partial U}{\partial V} \right)$$

$$T = VR * -P$$

$$PV = RT$$

